MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY, SIGHT CURRENT GENERATOR

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-60898 comprise the complete requirements for the Sight Current Generator Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-60898	Circuit Card Assemblies, General Specification for

DRAWINGS

U. S. Army, Frankford Arsenal

F10554188	Electronic Component Assembly, Sight Current Generator

PACKAGING DATA SHEET

10554188	Packaging of Circuit Card Assembly, Sight Current Generator

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 4931

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3. REQUIREMENTS

3.1 Fabrication, The assembly shall be manufactured in accordance with drawing F10554188 and drawings pertaining thereto.

3.2 General specifications. The assembly shall meet the requirements of MIL-C-60898.

3.3 Enviromental service conditions. See MIL-C-60898.

3.3.1 Performance at temperature extremes. For the conditions of 3.3.2 of MIL-C-60898, the requirements of 3,4 shall be met with exceptions as specified in the following subparagraphs,

3*3*1,1 Pin 19 output, There shall be -8,42 ± 0,50 vdc between pin 19 and pin 28.

3.3.1.2 Pin 23 output The voltage present between pin 23 and pin 28 shall be +15.4 ± 1.5 vdc.

3.3.1.3 Pin 9 output. Apply -1.1 vdc (rather than -1.25 vdc) and -1.7 vdc (rather than -1.55) between pin 12 (-) and pin 28 for the outputs of 3.4.4.

3.3.1.4 Pin 17 output. The voltage drop between pin 17 (+) and pin 28 shall be +21.0 ± 0.5 vdc and +1.2 ± 0.4 vdc.

3.3.1.5 Diodes, reverse current. There shall be 0.1 milliampere or less between the pins specified in Table 1.

3.3.1.6 Diodes, forward drop There shall be 1 ± 0.5 vdc between the pins specified in Table 11.

3.4 Electrical performance.

3.4.1 Conditions of operation. The requirements of 3.4.2 through 3.4.5 shall be met with the conditions shown in Figure 1, unless otherwise specified,

3.4.2 Pin 19 output. there shall be -8.42 ± 0.15 vdc between pin 19 (-) and pin 28.

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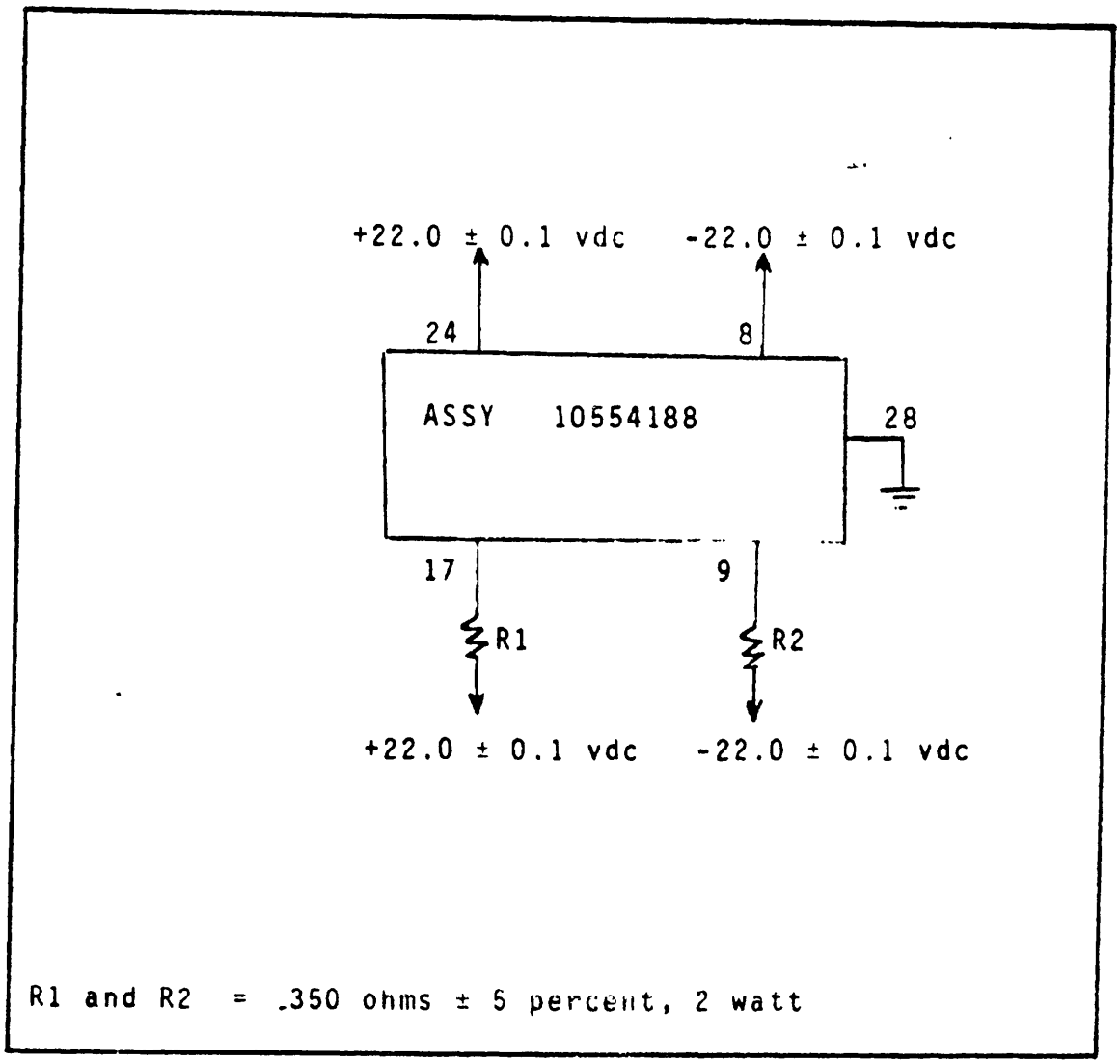


Figure 1

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3.4.3 Pin 23 output. There shall be +15.4 ± 1.0 vdc between pin 23 (+) and pin 28. With 28.0 ± 0.1 vdc between pin 24 (+) and pin 28, there shall also be +15.4 ± 1.0 vdc between pin 23 (+) and pin 28.

3.4.4 Pin 9 output. With-1.25 vdc applied between pin 12 (-) and pin 28, there shall be -21 ± 1 vdc between pin 9 (-) and pin 28. With -1.55 vdc applied between pin 12 (-) and pin 28, there shall be -0.3 ± 0.2 vdc between pin 9 (-) and pin 28.

3.4.5 Pin 17 output. The voltage between pin 17(+) and pin 28 shall be +21.0 ± 0.5 vdc. It shall remain at +21.0 ± 0.5 vdc when the voltage shown in Figure 2 is applied between pin 20 (+) and pin 28. The voltage shall go to +1.2 ± 0.4 vdc when +22.0 ± 0.1 Vdc is applied between pin 20 (+) and pin 28. When the applied +22.0 ± 0.1 vdc at pin 20 is removed the voltage shall remain +1.2 ± 0.4 vdc.

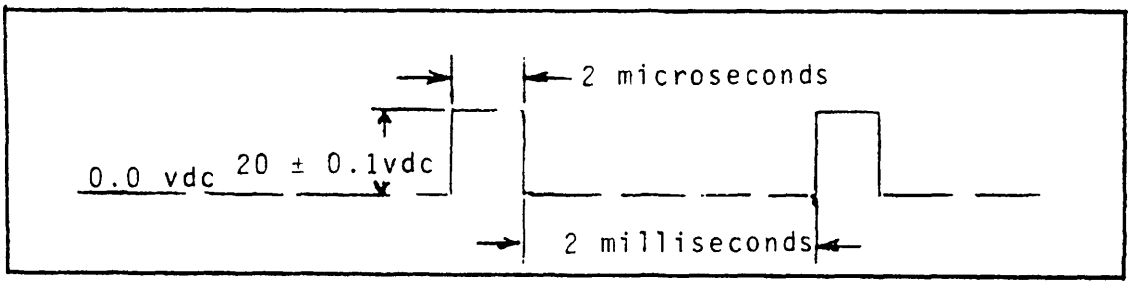


Figure 2

3.4.6 Diodes, reverse current. With +28.0 ± 0.1 vdc applied, respectively, between each of the two pins specified in Table I, there shall be 0.02 milliamperes or less current between these two pins.

Table I

28.0 ± 0.1 vdc - Polarity	
Pin (+)	Pin (-)
7	28
26	21
26	22

3.4.7 Diodes, forward drop, With a current of 1.0 ± 0.1 amperes applied between each of the two pins specified in Table II, there shall be 0,9 ± 0,3 vdc between these two pins.

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Table II

Ampere - Polarity	
Pin (+)	Pin (-)
28	7
21	26
22	26
28	24

3.4.8 Resistance. The resistance between pin 16 and pin 18 shall be 2.0 ohms (± 10 percent).

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4. QUALITY ASSURANCE PROVISIONS

4.1. General. The quality assurance provisions of MIL-C-60898 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. As specified in MIL-C-60898.

4.3 Control test sample. As specified in MIL-C-60898.

4.4 Functional tests. The following characteristics shall be 100 percent tested in accordance with the applicable test method.

Characteristic	Requirement	Procedure
100 Conditions of operation	3.4.1	4.5.2.1
101 Pin 19 output	3.4.2	4.5.2.2
102 Pin 23 output	3.4.3	4.5.2,3.
103 Pin 9 output	3.4.4	4.5.2.4
104 Pin 17 output	3.4.5	4.5.2.5
105 Diodes, reverse current	3.4.6	4.5.2.6
106 Diodes, forward drop	3.4.7	4.5.2.7
107 Resistance	3.4.8	4.5.2,8

4.4.1 Inspection equipment. As specified in MIL-C-60898 and 4.4.1.1.

4.4,1.1 Test equipment, In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4,5,

Item	Description
1. POWER SUPPLY	+22.0 ± 0.1 vdc nominal voltage with 500ma ; or greater, ripple less than 100 mv, voltage adjustable to +30.0 vdc
2. POWER SUPPLY	-22.00 ± 0.10 vdc with 500ma; or greater, ripple less than 100 mv
3* POWER SUPPLY	0 - 30 vdc, O - 1.5 ampere; current and voltage adjustable; ripple less than 1 percent of output.

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4	PULSE GENERATOR	0.5 microsecond to 0.5 millisecond adjustable pulse width adjustable duty cycle from 1 x 10^{-4} to 0.5. Adjustable amplitude from 0.1 to 20 volts into an output impedance of 600 ohms or less. Pulse rise and fall times, less than 0.2 microseconds each.
5.	OSCILLOSCOPE	Voltage and time accuracy and resolution of ± 5 percent or better, Rise time: 102 nanoseconds or less; sweep periods from 1 microsecond to 1 second per centimeter. Amplitude scales from 0.01 to 10 volts 1 centimeter, automatic triggering.
6,	DIGITAL VOLT-OHMETER	Accuracy and resolution of ± 0.2 percent or better from 0.9 to 99 vdc and accuracy and resolution of 1 percent or better from 1 to 100 kilohms. Input impedance of 10 megohms or greater on all d-c voltage scales.
7.	AMMETER	O-5 ampere with 2 percent resolution and accuracy at 0.02 milliampere and 1.0 amperes.
8.	RESISTORS	Two resistors, 350 ohms and ± 5 percent, 5 watt, each.
9.	OHMMETER	

4.5 Test methods and procedures.

4.5.1 Environmental. As specified in MIL-C-6089S except the requirements of 3.3.1 herein shall be verified during temperature extremes.

4,5.2 Electrical tests.

4.5.2.1 Conditions of operation. Unless otherwise specified all subsequent tests will be performed with the conditions of operation of 3.4.1 and Figure 1, Referring to Figure 1, connect RESISTORS R1 and R2, using the RESISTORS, Item 8 of 4.4.1.1.

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Connect POWER SUPPLIES, Items 1 and 2 of 4.4.1,1, for +22.0±0.1 and -22.0 ± 0.1 vdc, respectively, as shown in Figure 1 with their return leads connected to pin 28. During subsequent tests, measure all voltages referenced to pin 28 using the DIGITAL VOLT-OHMMETER, Item 6 of 4.4.1.1.

4.5,2.2 Pin 19 output Measure the d-c voltage at pin 19; verify conformance with 3.4.2.

4.5.2.3 Pin 23 output. Measure the d-c voltage at pin 23; verify conformance with 3.4.3, Increase the POWER SUPPLY, Item 1 of 4,4.1.1, voltage to 28,0 ± 0,1 vdc and again verify conformance with 3.4.3.

4,5.2.4 Pin 9 output. Set POWER SUPPLY, Item 3 of 4.4.1.1, to 1.25 vdc. Connect the negative output of the supply to pin 12 and the positive terminal-to pin 28". Verify that the voltage on pin 9 is -21,0 ± 1,0 vdc. Readjust the POWER SUPPLY to 1.55 vdc. Verify that the voltage on pin 9 is -0.3 ± 0.2 vdc.

4.5.2.5 Pin 17 output. Measure the d-c voltage on pin 17. Verify that the voltage is 21.0 ± 0.5 vdc. Connect the PULSE GENERATOR, Item 4 of 4.4.1.1, between pin 20(+) and pin 28. Connect the OSCILLOSCOPE, Item 5 of 4.4.1.1, vertical input between pin 20(+) and pin 28. Set up the PULSE GENERATOR to produce the pulse of Figure 2. Verify that the voltage on pin 17 remains at 21.0 ± 0.5 vdc. Disconnect the PULSE GENERATOR and apply 22.0 ± 0.1 vdc between pin 20(+) and pin 28. Using the POWER SUPPLY, Item 3 of 4.4,1.1, verify that the voltage on pin 17 is +1.2 ± 0.4 Vdc, Remove POWER SUPPLY, Item 3, and verify that the voltage on pin 17 remains at +1.2 ± 0.4 vdc.

4.5.2.6 Diodes reverse current, Connect the AMMETER, Item 7 and POWER SUPPLY, Item 3 of 4.4.1.1, per Figure 3 and apply the leads to each set of pins of Table I.

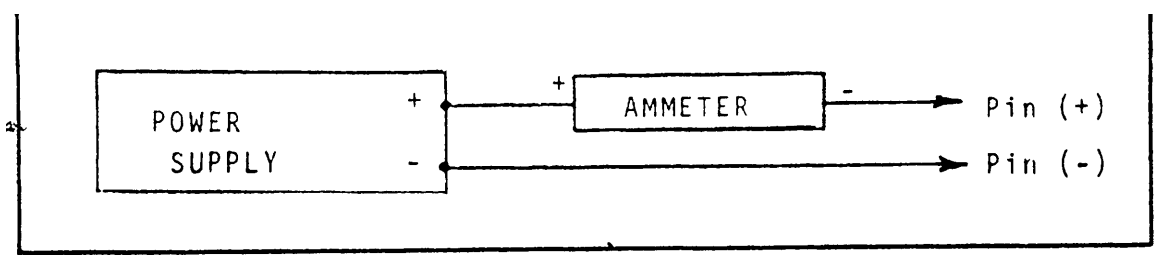


Figure 3

with the "POWER-SUPPLY SET AT 28.0 ± 0.1 vdc, verify that the AMMETER reads 0,02 MILLIAMPERE or less for each connection of Table I.

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4.5.2.7 Diodes, forward drop. Connect the AMMETER, item 7 of 4.4.1.1, and the POWER SUPPLY, item 3 of 4.4.1.1, Per Figure 3 and apply to each set of pins of Table II, With the POWER SUPPLY adjusted for an AMMETER reading of 1.0 ± 0.1 AMPERE, measure the d-c voltage between the two printed current board pins of each step. Using the DVOM, item 6 of 4.4.1.1, verify that this voltage is 0.9 ± 0,3 vdc.

4.5.2.8 Resistance, Connect the OHMMETER, item 9 of 4.4.1.1, between pin 16 and pin 18, verify the resistance of 3,4.8.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with Packaging Data Sheet 10554188.

6. NOTES

6.1 The notes specified in MIL-C-60898 are applicable to this specification.

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